Section 1: Executive Summary & Business Problem

1.1 Executive Summary

Credit risk assessment helps a loan officer decide whether to approve or reject a loan, using predictions from machine learning models such as Random Forest. Using a dataset of 32,581 records, multiple predictive models were built and evaluated, and the final system was deployed as an interactive Streamlit web application.1.2 Problem Statement & Objectives

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Traditional loan approval can take a long time, and with the significant growth in population in recent years, processing every request manually has become slow and difficult. The goals of this system are to:

- Classify the applicant's default probability with high predictive accuracy.
- Identify the key factors that drive each decision the model makes.
- Score the risk and recommend one of three outcomes: approve, reject, or manual review.

Section 2: Data Preprocessing Methodology

2.1 Dataset Description

The dataset has 32,581 applicant records across 12 features, including applicant age, annual income, employment history, loan amount, interest rate, loan grade, home ownership status, loan intent, and prior default history.

2.2 Data Cleaning & Feature Engineering

- Missing values: Every column was checked for missing values, and nulls in the affected columns were replaced with the median.
- Outliers: The data was then checked for outliers, which were replaced with median values to prevent skewness during model training.
- Categorical encoding: Ordinal loan grades (A to G) were mapped to numerical ranks (1 to 7), and prior default records were mapped to binary values (Y/N → 1/0). One-hot encoding was applied to the nominal attributes person_home_ownership and loan_intent.

Section 3: Exploratory Data Analysis (EDA) & Risk Findings

3.1 Income vs. Default

- Applicants who defaulted tend to earn less. The median annual income of defaulters is about 40,000, compared with about 60,000 for applicants who repaid. However, the two groups overlap considerably, so income alone cannot decide an application. Low income calls for a closer look, but it should be weighed together with the other factors.

3.2 Loan Intent and Default Rate

- Default rates differ clearly by loan purpose. Debt consolidation (29%), medical (27%) and home improvement (26%) loans default most often. Personal loans sit in the middle at 20%, while education (17%) and venture (15%) loans are the safest purposes. This suggests that higher-risk categories may need closer manual review or different interest rates.

3.3 Loan Amount Distribution

- Most loan requests fall between 5,000 and 15,000. Requests above about 20,000 are rare, so the model has seen fewer similar cases and its score is less certain for large loans.

3.4 Risk Across Loan Grades

- Loan grade is a strong indicator of risk. Applicants in grades A to C mostly repaid their loans. From grade D onwards, defaulters outnumber those who repaid, so grades D to G should be treated as high risk.

3.5 Credit History Length

- Most applicants have only 2 to 4 years of credit history, so most defaults also come from this group. Defaults become much less common for applicants with longer histories. A short credit history gives less proof of repayment behaviour, which makes these applicants harder to assess.

Section 4: Machine Learning Model Development & Performance Benchmarks

4.1 Model Selection & Training

A Random Forest Classifier was chosen for its ability to capture non-linear interactions among financial parameters. To handle class imbalance (safe loans greatly outnumbered defaults), class_weight='balanced' was applied.

4.2 Performance Metrics

The model achieved an exceptional **ROC-AUC score of ~0.93**, demonstrating a powerful ability to correctly separate prime borrowers from high-risk defaults. The classification report confirms high precision and recall across both classes, minimizing false approvals for subprime applicants.

Section 5: Deployment, System Architecture & Underwriting Framework

5.1 Architecture & Interactive Dashboard

The Credit Risk Assessment platform is deployed on Streamlit Cloud and backed by a GitHub repository for version control. The architecture uses a multi-view modular design with a sidebar navigation system, a login mechanism for secure session tracking, and a table of past records.

5.2 Theme-Adaptive UI & Visualizations

The website adapts to both light and dark themes. The exploratory data analysis charts, including boxplots, count plots, and distribution plots, adjust automatically to the theme the user selects.

5.3 Automated Explainability & Decision Workflow

Every evaluation triggers an instant risk calculation, followed by a “Why This Score” breakdown. The system classifies applicants into three actionable risk tiers:

- **Approved / Prime (probability < 30%):** Low-risk profile with a favorable Debt-to-Income (DTI) ratio.
- **Manual Review Needed (30% to 60%):** Borderline risk that needs a second review by an underwriter.
- **Rejected / Subprime (probability > 60%):** High default risk, typically due to a high DTI or an adverse credit history.